Md Tanzeel Adam Khan

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M.Tech (AI) from IIT Patna with research in LLM reasoning and search algorithms. 9 months building production AI systems at Mercedes-Benz R&D India (AMG SSV team). Skilled in multi-agent architectures, model fine-tuning (LoRA/PEFT), and retrieval-augmented generation.

Education

Indian Institute of Technology Patna 2024 – 2026
Master of Technology in Artificial Intelligence (CGPA: 8.75/10) *Patna, India*

Haldia Institute of Technology 2020 – 2024
Bachelor of Technology in CSE – AI & ML (CGPA: 8.9/10) *Haldia, India*

Publication

When Should Language Models Search? A Topology-Aware Analysis | *Submitted to ACL*

- Co-Author | Proposed A* search that explores multiple reasoning paths an LLM can take, with formally proven admissible heuristic ($h(s) \leq h^*(s)$). Evaluated across **8 benchmarks** $\times$ **3 scales (0.5B–14B)** $\times$ **4 methods** = 96 configurations, 500 instances each.
- Designed a zero-cost Topology Router (no LLM inference needed) using 14 structural features; **closes 40% of the gap** between the best single method and a perfect oracle. **95% routing accuracy** on CodeContests. Key feature predicts method success at $R^2 = 0.94$.

Research Experience

Indian Institute of Technology Patna Jul 2024 – Jun 2026
M.Tech Thesis | Advisor: Prof. Asif Ekbal | LLM Reasoning & Search *Patna, India*

- Implemented A* search with true branching (K=3 parallel LLM calls), self-consistency voting, entity-coverage heuristic, and repetitive output filtering. Designed 4-bucket evaluation framework (both-correct / CoT-only / A*-only / both-wrong) with **95% Wilson confidence intervals**. Ran **48,000+** paired evaluations with McNemar significance tests.
- Trained **LoRA adapters (rank-16, PEFT)** on Qwen-0.5B and LLaMA-3.1-8B to distill successful A* traces into linear CoT rationales. Built end-to-end pipeline: rationale rewriting, quality filtering, and SFT with HuggingFace Trainer. Achieved **33% rescue rate** on CoT failures with **50% preservation** of correct answers.
- Engineered a routing pipeline predicting best reasoning strategy from problem structure alone: 13 features, multi-classifier optimization (XGBoost, 5-fold CV, 100 seeds). Discovered Fluency-Correctness Asymmetry: **38%** of fluent traces are incorrect — explaining why probability-based routers fail.

Professional Experience

Mercedes-Benz Research & Development India (MBRDI) Nov 2025 – Aug 2026
AI Engineer Intern | AMG System Software Validation Team *Bengaluru, India*

- **iFAST (Intelligent Failure Analysis through Signal Tracing):** Architected an autonomous AI agent with 14 custom tools via MCP protocol, integrating **Gemini 2.5 Pro, GPT-5, and Claude** backends via AWS Bedrock. Performs probabilistic root-cause analysis fusing evidence from 8 independent sources using calibrated Bayesian scoring (Log-Likelihood Ratios). Achieved **54.8% time savings** and **97% accuracy** across 67K+ signals and 1,128 modules. Adopted by 2+ engineering teams.
- Built source-code dependency graph engine extracting 1.6M+ inter-module relationships via pattern-based static analysis across 3 repos. Signal-level verification eliminates **75% of false positives**. Replaced manual export (35% incomplete) with live git-based pipeline.
- **RAG Chatbot + KPI Dashboard:** Deployed hybrid BM25+ANN retrieval with cross-encoder reranking and LangGraph orchestration across 3 enterprise data sources (issue trackers + wiki). **77% answer rate**, **5.5$\times$ latency reduction** (22s $\rightarrow$ 4s) via semantic caching and async parallelism.
- **ASPICE SPOT Check (SWE.6):** Architected A2A system (JSON-RPC 2.0) for automated process quality assessment across **60+ evaluation dimensions** with mandatory source-citation, hierarchical scoring, and multimodal ingestion (7 formats).

Moshi Moshi Jun 2025 – Aug 2025
AI Engineer Intern *Bengaluru, India*

- Built automated lead generation pipeline (Selenium + open-source LLMs + SMTP cold emails), automating $\sim$**50%** of sales workflow. Developed **Inspira** – GenAI tool for designers; **35%** faster ideation, **25%** faster handoff.

Technical Skills

ML/DL: PyTorch, HuggingFace Transformers, LoRA/QLoRA/PEFT, XGBoost, scikit-learn, Bayesian Inference
LLM Systems: LangGraph, LangChain, RAG, MCP Protocol, A2A/ReAct Agents, Tool-Use, vLLM, Ollama
Retrieval: BM25, ANN, Cross-Encoders, Reciprocal Rank Fusion, ChromaDB, LanceDB, Sentence-Transformers
Cloud & Infra: AWS (Bedrock), Docker, Git, CI/CD, FastAPI, Streamlit, PySide6/Qt, API Design
Engineering: Python, asyncio, SQL, NumPy, Pandas, SciPy, NetworkX, Distributed Systems, Event-Driven

Additional

Leadership: Designed & conducted "AI IN LOOP" workshop for MBRDI RD IDA team – upskilling 30+ engineers on LLM agents, RAG systems, and prompt engineering
Teaching: Teaching Assistant – Deep Learning & AI courses, IIT Patna